

Nested Chain Rule Derivatives Assignment

Directions: Assume f and g are differentiable. Find the derivative of each function with respect to x .

1. $y = f(3x^2 - 2x)$

2. $y = g(\sin(4x))$

3. $y = \cos(f(x^2 + 1))$

4. $y = f(g(2x^3 - x))$

5. $y = \sin(g(\cos(3x)))$

6. $y = g(\cos(f(2x^2 - 5x)))$

7. $y = f(\sin(g(x^2 - 4x)))$

8. $y = \tan(f(\sqrt{x^2 + 4}))$

9. $y = g(\sin(f(\cos(2x))))$

10. $y = f(\cos(g(\sqrt{x^2 + \sin(x)})))$

11. $y = \sin(g(\tan(f(x^3 - \cos(x)))))$

12. $y = g(\cos(f(\sin(\sqrt{x^2 + 3x}))))$

13. $y = f(\sin(g(\cos(\sqrt{x^4 - \tan(x^2)}))))$

14. $y = g(\tan(f(\sin(\sqrt{x^3 - \cos(2x^2)}))))$

15. $y = \cos(\sin(g(\tan(\sqrt{x^2 + f(x^3)}))))$